

RICARDO ALEXANDER BARTRA QUISPE

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PROFESSIONAL SUMMARY

AI and backend engineer focused on document intelligence, OCR validation, microservice architectures, and applied machine learning. Built production-oriented prototypes for Peruvian identity and education documents, human-in-the-loop OCR review, SAT practice workflows, legal/regulatory scraping, and image-recognition APIs. Strong practical stack across Python/FastAPI, Rust/Axum, Java/Spring Boot, TypeScript/Next.js, PostgreSQL, Redis, Docker Compose, Prisma, SQLAlchemy, OCR engines, PDF parsing, computer vision, and metrics-driven evaluation.

SKILLS

Languages: Python, Rust, Java, TypeScript, JavaScript, SQL, Bash

Backend: FastAPI, Axum, Express, Spring Boot, REST APIs, async services, JWT, webhooks, service orchestration, API gateways

AI, OCR & Data: OCR, document understanding, computer vision, OpenCV, Docling, RapidOCR, GLM-OCR/Ollama, YOLO, pdfplumber, PyMuPDF, pandas, NumPy, SciPy, embeddings, RAG concepts

Infrastructure: Docker, Docker Compose, PostgreSQL, Redis Streams, SQLite, MySQL, S3 storage, Prometheus, GitHub Actions

Frontend: Next.js App Router, React, Tailwind CSS, Radix UI, TanStack Table, React Hook Form, Zod, NextAuth, Prisma

Evaluation & Optimization: Benchmarking, extraction metrics, quality gates, fuzzy matching, regression analysis, model/service latency, ACO, PSO, ABC, combinatorial optimization

EXPERIENCE

AI / Backend Engineering Contributor [Start Date] – Present
Independent, Academic and Project-Based Work Lima, Peru

- Built OCR and document-validation services for DNI, COE, CLA, CNR, FRP and related document families using Python, FastAPI, OCR engines, PDF parsers, image preprocessing, fuzzy matching, and structured JSON outputs.
- Designed a Docker Compose backend for document processing with Rust APIs, Python workers, PostgreSQL, Redis queues, review services, rule configuration, status tracking, authentication, rate limiting, and observability endpoints.
- Developed Rust services with Axum, Tokio, Redis, JWT, Prometheus, tracing, and SQL-backed service boundaries for gateway, ingestion, queue dispatching, rules, and status APIs.
- Implemented human-in-the-loop review workflows with FastAPI, SQLAlchemy async, Alembic, Pydantic, PostgreSQL, S3-compatible storage, and manual QA states for OCR pipelines.
- Created OCR benchmarks and validation reports measuring extraction quality, image resolution impact, execution time, and field-level match quality across document formats.
- Built web application workflows with Next.js, React, TypeScript, Prisma, NextAuth, Tailwind, Radix UI, Zod, and TanStack Table for role-based portals, assignments, testing flows, and administrative dashboards.

Research Assistant – Applied AI and Optimization [Start Date] – Present
[University / Research Lab / Project Name] Lima, Peru

- Designed experiments for applied AI, document understanding, retrieval-augmented generation, model evaluation, quantization, and lightweight deployment.
- Implemented and compared metaheuristic approaches including Ant Colony Optimization with Time Windows, Artificial Bee Colony, and Particle Swarm Optimization for routing, scheduling, and constrained search problems.
- Built reproducible evaluation loops with train/test separation, metric computation, confusion matrices, regression metrics, custom objective functions, and comparative experiment summaries.
- Produced technical documentation, architecture notes, implementation guides, API references, and benchmark reports for AI and backend systems.

SELECTED PROJECTS

OCR Document Processing Platform Rust, Python, FastAPI, Axum, PostgreSQL, Redis, Docker

- Designed a multi-service OCR backend with API gateway, ingest API, status API, rules API, queue service, review API,

and specialized detect/extract workers.

- Implemented document-family workflows for DNI, CNR, COE, FRP, CLA and LN using OCR, layout detection, PDF parsing, validation rules, and worker-based processing.
- Used Redis Streams for dispatching, PostgreSQL for service-owned persistence, Docker Compose for orchestration, Prometheus for metrics, and JWT/internal tokens for service access.
- Added human review integration for low-confidence or manually validated OCR results.

DNI and RENIEC Document Validators

Python, FastAPI, OpenCV, Docling, GLM-OCR, Ollama

- Built a DNI validation pipeline with image quality checks, resizing, OCR by text boxes, dual-format field extraction, exact DNI matching, fuzzy name matching, and JSON/HTML reporting.
- Supported multiple OCR paths including GLM-OCR through Ollama and Docling, with benchmarking for resolution, latency, and extraction quality.
- Extended document extraction patterns to RENIEC and related Peruvian document validation workflows using structured outputs and test datasets.

COE Certificate Extraction and Validation

Python, FastAPI, pdfplumber, pandas, NumPy

- Built extraction logic for Peruvian Certificates of Studies across DCN 2009, CNEB 2016, and transition formats.
- Combined table extraction, regex-based text parsing, duplicate consolidation, invalid-table detection, structured JSON generation, and Markdown/CSV metric reports.
- Exposed the pipeline through a REST API and standalone benchmark scripts for batch processing and timing analysis.

SAT Practice Platform

Next.js, React, TypeScript, Prisma, NextAuth, Tailwind

- Developed a role-based SAT preparation platform for students, teachers, admins, and parents with protected portals and validation states.
- Implemented assignment flows, timed practice attempts, autosave, score submission, explanations, full practice tests, user invitations, password reset, and parent consent workflows.
- Used Prisma with SQLite, NextAuth credentials/JWT, React Hook Form, Zod validation, Radix UI components, TanStack Table, KaTeX, and email notifications.

Legal Researcher for Peruvian Telecom Regulation

Node.js, Express, SQLite, Cron, PDF processing

- Built a daily scraper for El Peruano and OSIPTEL legal norms related to telecommunications, consumer protection, mobile services, SIM cards, and regulatory projects.
- Implemented source-specific filters, PDF download, SHA-256 tracking, SQLite persistence, scheduled execution, HTTP API access, email reports, and webhook delivery.
- Wrote automated tests using Node's native test runner and kept source keywords centralized for maintainable regulatory matching.

Image Recognition REST and Scheduler APIs

Java, Spring Boot, MySQL, S3, JWT

- Implemented REST APIs for image recognition workflows, verification rules, users, security, image formats, scheduler operations, and external-system integrations.
- Used Spring Boot, Spring Security, JPA, validation, OpenAPI documentation, JWT authentication, MySQL, S3 integration, Apache POI, and layered endpoint documentation.

Optimization and Model Evaluation Experiments

Python, Java, ML, Metaheuristics

- Implemented ACO-TW, ABC, PSO, gradient-descent-based reasoning, and custom experiment loops for routing, assignment, scheduling, and parameter-search problems.
- Evaluated models and algorithms with custom objective functions, classification/regression metrics, repeated runs, convergence analysis, and hyperparameter sensitivity.

EDUCATION

Pontificia Universidad Catolica del Peru

[Start Date] – [End Date]

Bachelor's Degree in Informatics Engineering / Computer Engineering

Lima, Peru

- Relevant Coursework: Artificial Intelligence, Software Engineering, Operating Systems, Information Security, Enterprise Architecture, ERP, Business Intelligence, Simulation, Statistics, Data Visualization.
- Research Focus: document intelligence, Spanish-language AI systems, RAG evaluation, OCR validation, model efficiency, and applied optimization.

CERTIFICATIONS AND TRAINING

- RAG, LLM evaluation, prompt engineering and lightweight AI deployment – applied project experience
- Docker, service orchestration, API gateways, queues and backend observability – applied project experience
- Machine learning, deep learning, OCR, computer vision and optimization – academic and project experience
- Hyperledger Fabric, ISO 20022 and digital asset infrastructure – independent specialization